Exp No. 9

20/8/2026

Implementation of Subnetting in Cisco Packet Tracer Simulator

a) Design multiple subnets with a suitable number of hosts

Aim:

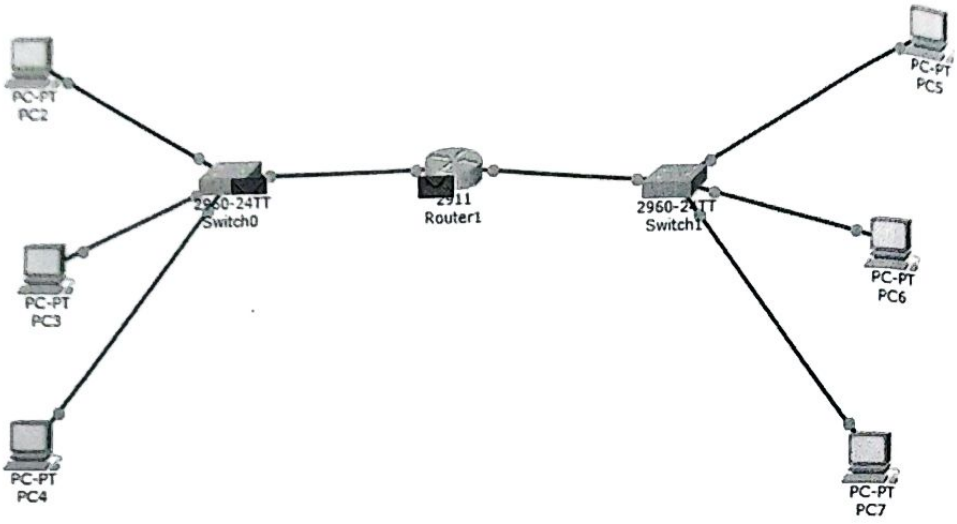
To implement subnetting by individual a large network into multiple smaller subnetworks according to the required number of hosts and to study the allocation of IP address in Cisco Packet Tracer

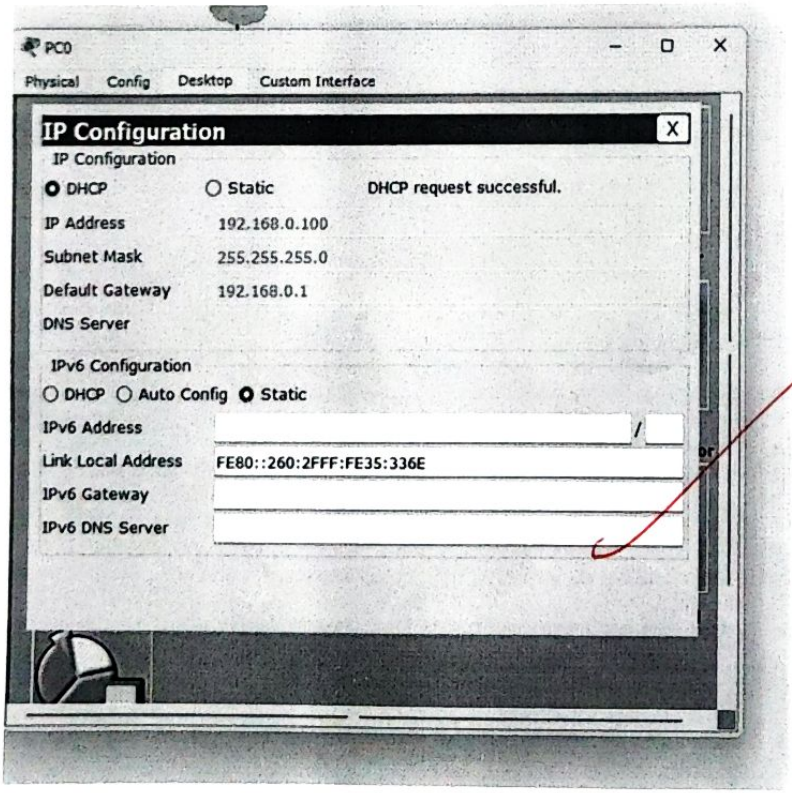
Steps:

1. Open Cisco packet tracer and create a new project
2. Design the network topology by placing the required network devices such as routers, switches and PCs.
3. Select a suitable network address for the experiment
4. Calculate the number of subnets required
5. Determine the number of hosts needed in each subnet
6. Calculate the subnet masks using subnetting techniques.
7. Divide the network into multiple subnets
8. Assign a seperate network address to each subnet
9. Verify the subnet calculations and host ranges.
10. Save the network topology.

Result:

Thus, the multiple subnets were successfully create according to the required number of ~~hosts~~ nodes and the network address was efficiently divided into smaller subnets.


PC-PT
PC2
PC-PT
PC5
2960-24TT
Switch0
2911
Router1
2960-24TT
Switch1
PC-PT
PC3
PC-PT
PC6
PC-PT
PC4
PC-PT
PC7


PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
DHCP request successful.
IP Address
192.168.0.100
Subnet Mask
255.255.255.0
Default Gateway
192.168.0.1
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::260:2FFF:FE35:336E
IPv6 Gateway
IPv6 DNS Server

b) Assign Static IP Address Across All subnets and Connect the subnets through a Router

AIM:

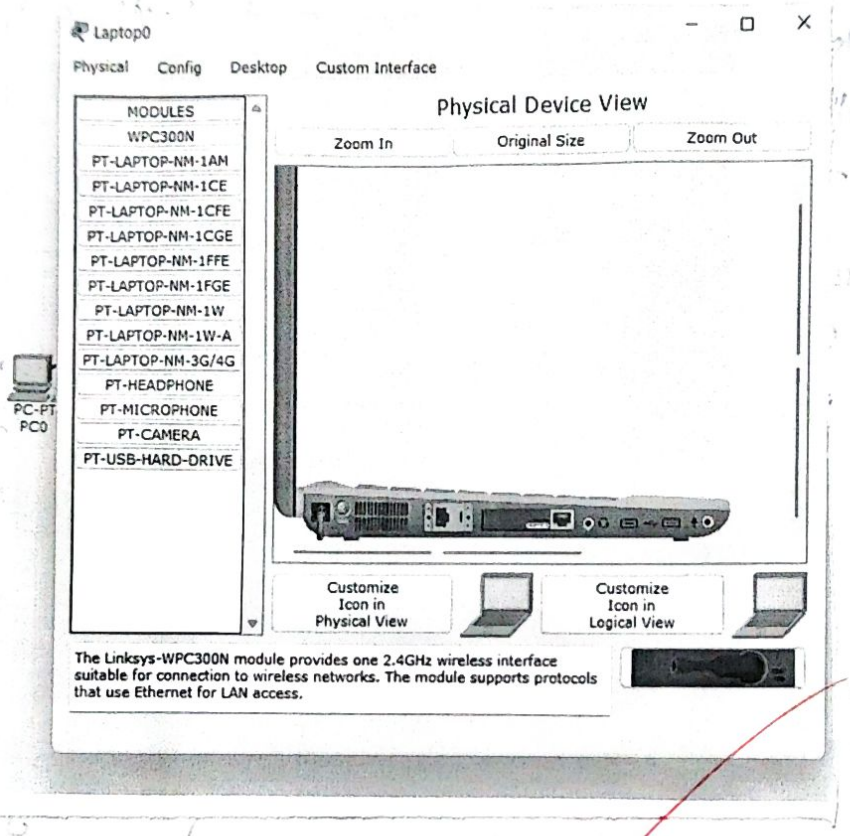
To assign static IP address to all devices to different subnets and establish communication between the subnets through a router.

STEPS:

1. Place a router in the network topology
2. Connect the router to the switches using appropriate cables
3. Configure the router interface
4. Assign and IP address and subnet mask to each router interfaces.
5. Configure the switches
6. Assign static IP address to all computers in every subnet.
7. Configure the subnet mask for each devices.
8. Assign the appropriate default gateway to each computer
9. Verify the IP configuration of all devices.
10. Save the router configuration.
11. Check the network topology for configuration errors.
12. Verify that all devices are properly connected.

RESULTS:

Thus, static IP address were successfully assigned to all devices and the communication between different subnets was established through the router.


Laptop0
Physical
Config
Desktop
Custom Interface
MODULES
WPC300N
PT-LAPTOP-NM-1AM
PT-LAPTOP-NM-1CE
PT-LAPTOP-NM-1CFE
PT-LAPTOP-NM-1CGE
PT-LAPTOP-NM-1FFE
PT-LAPTOP-NM-1FGE
PT-LAPTOP-NM-1W
PT-LAPTOP-NM-1W-A
PT-LAPTOP-NM-3G/4G
PT-HEADPHONE
PT-MICROPHONE
PT-CAMERA
PT-USB-HARD-DRIVE
Physical Device View
Zoom In
Original Size
Zoom Out
Customize Icon in Physical View
Customize Icon in Logical View
The Linksys-WPC300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.
PC-PT
PC0

c) Simulate packet Transmission Across the Subnets and observe the result

9) When subnets are connected through a Router

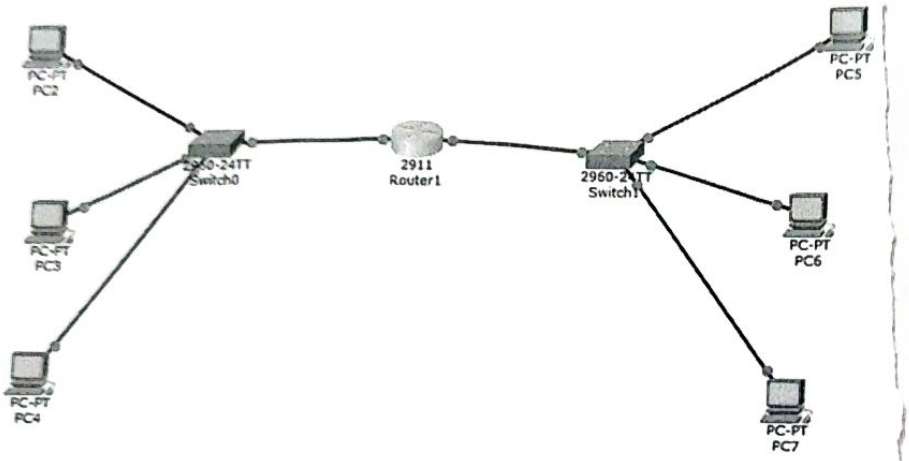
AIM: To analyze packet Transmission between different subnets when a router is used for communication

STEPS:

1. Open Simulation mode in Cisco packet Tracer
2. Select a source devices from one subnet
3. Select a destination devices from another subnet
4. Generate a packet using the simple PDU tool.
5. Observe the movement of the packet through the network
6. Analyze how the router ~~was~~ receive the packet
7. Observe the forwarding process performed by the router.
8. Verify the packet transmission between different subnets
9. Use the ping ~~request~~ command to test connectivity
10. Observe the successful delivery of packets.
11. Record the simulation results.
12. Analyze the routing process

RESULT:

Thus, packets were successfully transmitted between different subnets through the router. The router received the packet from the source subnet, processed them and forward them to the destination subnet successfully ping response confirmed paper communication between all network devices.




```
        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no


Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
```

9) When subnets are not connected through a Router

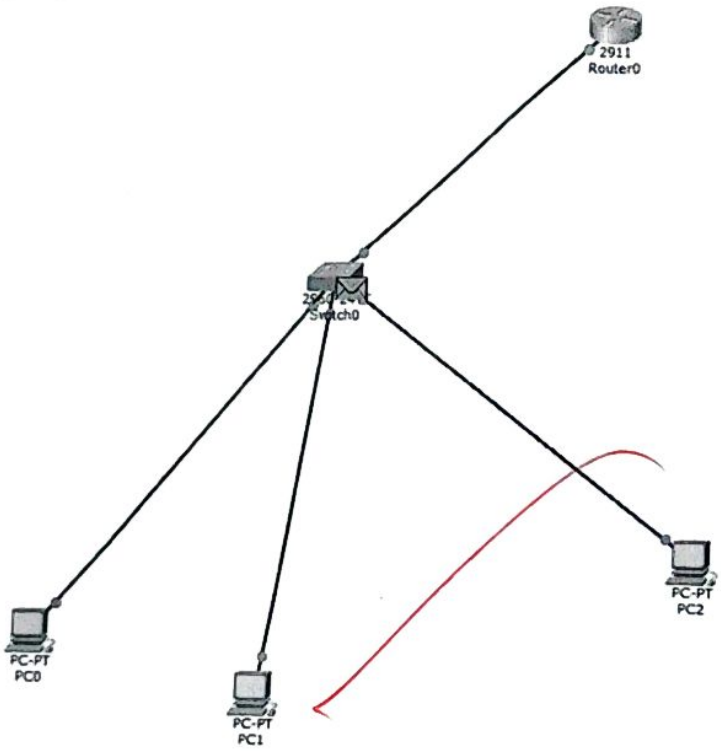
Aim: To analyze packet transmission between different subnets when no router is available

STEPS:

1. Remove the router from the network topology
2. Open simulation mode.
3. Select devices belonging to different subnets.
4. Generate packets using the simple PDU tools
5. Observe the packet transmission
6. Perform a ping test between devices located in different subnets.
7. Analyze the packet flow.
8. Observe the simulation events.
9. Identify the point at which packet transmission fails.
10. Compare the result with the network that uses a router
11. Record the observations.
12. Analyze the importance of a router in inter-network communication.

RESULT:

Thus, communication between devices located in different subnets failed when a router was not used. The packets could not reach the destination because there was no routing device to forward them. This experiment demonstrated the important of routers in enabling communication between different subnetworks.


2911
Router0
Switch0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2